Year 9 Mathematics

Unit 1: Number

1.4 Standard Form

Standard Form

STARTER · 5 MIN

Date: _____ **Title: Standard Form**

LEARNING OBJECTIVES

- Write large numbers in standard form
- Write small numbers in standard form
- Order numbers written in standard form

STARTER *Answer in your book. No calculators.*

1. Work out 2.52×100
2. Work out $3.65 \div 100$
3. Write 10^4 as a number
4. Write 10^{-3} as a decimal

Starter — Answers

STARTER

Tick or correct in green pen

1. **252**
2. **0.0365**
3. **10 000**
4. **0.001**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can write large numbers in standard form	☐
SILVER	I can write small numbers in standard form	☐
GOLD	I can order numbers written in standard form	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Standard Form

INTRODUCTION · 3 MIN

Standard form is a short way to write very large or very small numbers.

The form

A number between 1 and 10, times a power of 10.

$A \times 10^{n}$, $1 \leq A < 10$

Large numbers

Positive power of 10.

$41\,000 = 4.1 \times 10^{4}$

Small numbers

Negative power of 10.

$0.0041 = 4.1 \times 10^{-3}$

Ordering

Compare the powers first, then A.

$3.2 \times 10^{5} > 9.8 \times 10^{4}$

Examples 1 and 2 — Standard Form to Ordinary Numbers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $A \times 10^n$, $1 \leq A < 10$ $\times 10^{-n}$ means $\div 10^n$

Example 1

Write 3.7×10^3 as an ordinary number.

Example 2

Write 9.3×10^{-3} as an ordinary number.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

× 10^3 means × 1000

3.7 × 1000

= 3700

Tip: the digits move 3 places left.

Example 2

× 10^{-3} means ÷ 1000

9.3 ÷ 1000

= 0.0093

Tip: a negative power gives a small number.

Task 1

PRACTICE · 9 MIN

BRONZE *Write as an ordinary number*

Hint $A \times 10^n$, $1 \le A < 10$ $\times 10^{-n}$ means $\div 10^n$

1. 2.5×10^4

2. 8.1×10^2

3. 5.4×10^7

4. 7.3×10^{-2}

5. 1.5×10^{-4}

★ 4.9×10^{-6}

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **25 000**
2. **810**
3. **54 000 000**
4. **0.073**
5. **0.000 15**

★ **0.000 004 9**

Examples 3 and 4 — Writing Numbers in Standard Form

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $A \times 10^n$, $1 \leq A < 10$ less than 1: negative power

Example 3

Write 312 500 000 000 in standard form.

Example 4

Write 0.000 52 in standard form.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

A = 3.125

3.125 × 100 000 000 000

= 3.125 × 10^{11}

Tip: count the places the digits move.

Example 4

A = 5.2

5.2 ÷ 10 000

= 5.2 × 10^{-4}

Tip: the 5 moves 4 places right.

Task 2

PRACTICE · 9 MIN

SILVER *Write in standard form*

Hint $A \times 10^n$, $1 \le A < 10$ less than 1: negative power

1. 59 000
2. 5323
3. 720 000 000
4. 0.0601
5. 0.000 000 072

★ 0.000 409

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 5.9×10^{4}

2. 5.323×10^{3}

3. 7.2×10^{8}

4. 6.01×10^{-2}

5. 7.2×10^{-8}

★ 4.09×10^{-4}

Examples 5 and 6 — Ordering Standard Form

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint compare the powers of 10 first, then A times larger: divide A, subtract powers

Example 5

(2 marks)

Write in order, smallest first.

2.1×10^{5}, 3.7×10^{-2},

9.4×10^{2}, 1.8×10^{5}

Example 6

(2 marks)

How many times larger is

6×10^{5} than 3×10^{2}?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Powers first: −2, 2, 5, 5

Same power 10^5: 1.8 < 2.1

3.7×10^{-2}, 9.4×10^{2},

1.8×10^{5}, 2.1×10^{5}

Tip: same power? Then compare A.

Example 6

Divide the numbers and the powers

$6 \div 3 = 2$, $10^5 \div 10^2 = 10^3$

2×10^3

= 2000 times

Tip: 5 − 2 = 3

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint compare the powers of 10 first, then A times larger: divide A, subtract powers

1 Which is larger?

3.1×10^{4} or 9.9×10^{3}

2 Which is larger?

2.4×10^{-3} or 2.4×10^{-2}

3 Which is smaller?

7×10^{-5} or 1×10^{-4}

4 Which is smallest?

4.2×10^{-2}, 2.44×10^{-1}, 4×10^{-1}

5 How many times larger is

8×10^{6} than 2×10^{3}?

★ How many times larger is

9×10^{-2} than 3×10^{-5}?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 3.1×10^{4}

2 2.4×10^{-2}

3 7×10^{-5}

4 4.2×10^{-2}

5 4000 times

★ 3000 times

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write 6.2×10^{-3} as an ordinary number.

SILVER Write 480 000 in standard form.

GOLD Which is smaller: 5×10^{-6} or 8×10^{-7}?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **0.0062**

SILVER **4.8×10^{5}**

GOLD **8×10^{-7}**

Exam Practice — Ordinary Numbers

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $A \times 10^n$, $1 \le A < 10$ $\times 10^{-n}$ means $\div 10^n$

Question 1

(2 marks)

Write as an ordinary number.

(a) 6.03×10^5 (1)

.....

(b) 8.4×10^{-4} (1)

.....

Question 2

(1 mark)

Light travels 9.46×10^{12} km in a year.
Write this as an ordinary number.

..... km

Exam Practice — Writing in Standard Form

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $A \times 10^n$, $1 \le A < 10$
less than 1: negative power

Question 3

(2 marks)

Write in standard form.

(a) 72 000 (1)

.....

(b) 0.000 506 (1)

.....

Question 4

(1 mark)

A dust particle is 0.000 000 25 m wide.
Write this in standard form.

..... m

Exam Practice — Ordering

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint compare the powers of 10 first, then A times larger: divide A, subtract powers

Question 5

(2 marks)

Write these numbers in order of size.
Start with the smallest.

3.4×10^{-2} 2.9×10^{-3}

3×10^{-3} 3.25×10^{-2}

.....

Question 6

(2 marks)

An atom's radius is 5.3×10^{-11} m.
Light's wavelength is 5.5×10^{-7} m.
Jamal says the wavelength is about 10 000 times bigger. Is he correct?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 603 000 B1

(b) 0.000 84 B1

Tip: negative power: the number is less than 1.

Question 2

9 460 000 000 000 km B1

Tip: the digits move 12 places left.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 7.2×10^{4} — B1

(b) 5.06×10^{-4} — B1

Tip: A must be at least 1 and less than 10.

Question 4

2.5×10^{-7} m — B1

Tip: the 2 moves 7 places right.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

group by power, then compare A — M1

2.9×10^{-3}, 3×10^{-3},

3.25×10^{-2}, 3.4×10^{-2} — A1

Tip: 10^{-3} is smaller than 10^{-2}.

Question 6

$10^{-7} \div 10^{-11} = 10^{4}$ — M1

Yes, $10^{4} = 10\,000$ — A1

Tip: 5.5 and 5.3 are about the same.